

Literature review and verified reference register

Scope and evidence limits

This concise background review covers edge/cloud placement, edge intelligence and resource-aware inference, anomaly-detection model families described in the project, and the specific project comparison. It draws on academic database records and the abstracts/passages saved for this draft (retrieved 2026-10-10). Open web access was not enabled, so no official repository, publisher landing page, or web-only technical documentation was inspected. Most publication evidence below is abstract-level; this is not a systematic review and should not be presented as exhaustive. A directly comparable primary study for the exact proposed three-architecture evaluation was not identified in the retrieved passages. That observation is restricted to this search and is not proof of nonexistence.

1. Edge and cloud computing

Edge computing moves computation toward the network edge and is motivated by response-time requirements, bandwidth cost, energy constraints, and data safety/privacy concerns [1]. Mobile-edge computing research describes moving computation, network control, and storage closer to resource-limited devices, with offloading and joint radio/computational resource management as prominent problems [2]. These motivations support the project's comparison of cloud transmission against local processing, but they do not guarantee a particular latency or resource gain: those outcomes depend on the workload and measurement boundary.

Edge intelligence literature extends the placement question to model inference. Zhou and colleagues survey edge-AI architectures and describe performance, resource, privacy, and communication concerns; they emphasize resource-aware model design and choices among model candidates [3]. The project's LIGHT/STANDARD/HEAVY model labels are consistent with the general engineering concern of choosing different model costs, but the labels and actual costs of its three specific models are project descriptions that require implementation-level benchmarking.

2. Anomaly-detection methods in the described stack

Isolation Forest is grounded in the observation that rare, different data points can be isolated efficiently. Liu, Ting, and Zhou describe the method as avoiding distance and density measures and using subsampling to achieve low linear time complexity and small memory requirements; their abstract also reports comparisons with several alternatives [4]. These are properties of the published algorithm and its reported study, not proof that the participant's implementation is correctly configured or is fastest on the project data.

For sequence data, Malhotra and colleagues describe an LSTM encoder-decoder that learns normal time-series behavior and detects anomalies using reconstruction error [5]. Their abstract reports evaluations on several public time-series datasets and real-world

engine data, across different temporal patterns. Hundman and colleagues describe LSTM-based anomaly detection on labeled spacecraft telemetry and a nonparametric dynamic thresholding method, highlighting false-positive mitigation as a practical component [6]. These works justify discussing sequence reconstruction and threshold calibration, but no retrieved source establishes that the described project trains these models as specified. The project-specific standard AutoencoderModel has no independently verified implementation or bibliographic model description in the supplied materials; its structure and role must be checked.

3. Adaptive and resource-aware inference

Taylor and colleagues present adaptive selection among pretrained deep models for embedded inference based on desired accuracy and inference time. The abstract reports an image-classification evaluation on an embedded platform and reports improvements over the most-capable single model in that setting [7]. This is evidence that adaptive model selection has prior research precedent, not evidence about anomaly-detection tasks or the submitted framework. Edge-intelligence surveys likewise frame model choice and resource use as a multi-metric systems question [3].

The submitted project summary describes a resource-aware utility mechanism choosing between local processing, transmission, and model classes. It reports that STANDARD was selected most often and that adaptive results match static edge detection. The conclusion supported by these facts is limited: the current reported run does not show a measured detection advantage from adaptation. To establish a contribution, the evaluation should expose controller decisions, define its utility and weights, and include resource/load scenarios in which choices differ from the static baseline.

4. Evidence synthesis and evaluation requirements

The literature supports the engineering motivation for edge placement and model/resource trade-offs, while the available anomaly papers establish distinct algorithmic precedents. These are separate evidence streams: a favorable edge-computing rationale does not itself establish a better anomaly detector, and an anomaly model's performance on another dataset does not establish performance on this synthetic experiment. Accordingly, the comparison must report both detection and systems measures with aligned populations, measurement boundaries, and explicit thresholds.

The supplied reported metrics show a recall/precision trade-off and poor recall for the two edge configurations; the same reported values for static and adaptive edge do not demonstrate adaptation benefit. The system results suggest lower byte counts for both edge methods, but latency cannot be compared directly because the supplied description assigns different timing boundaries to centralized and edge results. Confusion matrices, class prevalence, threshold details, repeated trials, consistent end-to-end timing, and the underlying source records are needed before stronger claims can be made. These are methodological deductions from the supplied results, not findings from the cited literature.

Verified reference register

The following references were matched to academic records or exact paper lookups and have stable identifiers. DOI/record details are listed for identification; final venue formatting should be checked against the official publisher record before publication. The works without venue information in the retrieved record are explicitly marked.

1. Shi, W., Cao, J., Zhang, Q., Li, Y., & Xu, L. (2016). **Edge Computing: Vision and Challenges**. *IEEE Internet of Things Journal*. <https://doi.org/10.1109/JIOT.2016.2579198>
2. Mao, Y., You, C., Zhang, J., Huang, K., & Letaief, K. B. (2017). **A Survey on Mobile Edge Computing: The Communication Perspective**. *IEEE Communications Surveys & Tutorials*. <https://doi.org/10.1109/COMST.2017.2745201>
3. Zhou, Z., Chen, X., Li, E., Zeng, L., Luo, K., & Zhang, J. (2019). **Edge Intelligence: Paving the Last Mile of Artificial Intelligence With Edge Computing**. *Proceedings of the IEEE*. <https://doi.org/10.1109/JPROC.2019.2918951>
4. Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2012). **Isolation-Based Anomaly Detection**. *ACM Transactions on Knowledge Discovery from Data*. <https://doi.org/10.1145/2133360.2133363>
5. Malhotra, P., Ramakrishnan, A., Anand, G., Vig, L., Agarwal, P., & Shroff, G. (2016). **LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection**. arXiv:1607.00148. <https://arxiv.org/abs/1607.00148>
6. Taylor, B., Sanz Marco, V., Wolff, W., Elkhathib, Y., & Wang, Z. (2018). **Adaptive deep learning model selection on embedded systems**. DOI: <https://doi.org/10.1145/3211332.3211336>. Venue as identified in retrieved record: Lancaster EPrints (Lancaster University); formal conference venue [TO BE VERIFIED].
7. Hundman, K., Constantinou, V., Laporte, C., Colwell, I., & Soderstrom, T. (2018). **Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding**. DOI: <https://doi.org/10.1145/3219819.3219845>. Venue metadata [TO BE VERIFIED].

Items requiring verification or not found

- No independently verified paper describing the exact `AutoencoderModel` implementation was identified from the supplied description; do not attach a guessed citation to it.
- No publication in the retrieved passages establishes this project's three-arm comparison or its results. This does not imply the broader literature lacks such studies.
- The retrieved literature search did not verify a primary reference specifically supporting metric definitions or a PR-AUC interpolation convention for this package. Definitions in the paper are conventional and the actual implementation must be checked in `MetricsCalculator`.

- The formal proceedings venue for Taylor et al. and the publication venue details for Hundman et al. should be verified against the DOI record before the final camera-ready bibliography.

References

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, 2016, doi: [10.1109/JIOT.2016.2579198](https://doi.org/10.1109/JIOT.2016.2579198).
- [2] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Communications Surveys & Tutorials*, 2017, doi: [10.1109/comst.2017.2745201](https://doi.org/10.1109/comst.2017.2745201).
- [3] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, "Edge intelligence: Paving the last mile of artificial intelligence with edge computing," *Proceedings of the IEEE*, June 2019, doi: [10.1109/jproc.2019.2918951](https://doi.org/10.1109/jproc.2019.2918951).
- [4] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation-based anomaly detection," *ACM Transactions on Knowledge Discovery from Data*, 2012, doi: [10.1145/2133360.2133363](https://doi.org/10.1145/2133360.2133363).
- [5] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, "LSTM-based encoder-decoder for multi-sensor anomaly detection," *arXiv*, 2016, Available: <https://arxiv.org/pdf/1607.00148v2>
- [6] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding," 2018, doi: [10.1145/3219819.3219845](https://doi.org/10.1145/3219819.3219845).
- [7] B. Taylor, V. S. Marco, W. Wolff, Y. Elkhatib, and Z. Wang, "Adaptive deep learning model selection on embedded systems," *Lancaster EPrints (Lancaster University)*, pp. 31–43, 2018, doi: [10.1145/3211332.3211336](https://doi.org/10.1145/3211332.3211336).